

RMO(INDIA) – 2013 Mumbai

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Let ABC be an isosceles triangle with $AB = AC$ and let Γ denotes its circumcircle. A point D is on arc AB of Γ not containing C. A point E is on arc AC of Γ not containing B. If $AD = CE$ prove that BE parallel to AD.
 2. Find all triples (p, q, r) of primes such that $pq = r + 1$ and $2(p^2 + q^2) = r^2 + 1$.
 3. A finite non empty set of integers is called 3-good if the sum of its elements is divisible by 3. Find the number of non empty 3-good subsets of $\{0, 1, 2, \dots, 9\}$.
 4. In a triangle ABC, points D and E are on segments BC and AC such that $BD = 3DC$ and $AE = 4EC$. Point P is on line ED such that D is the midpoint of the segment EP. Lines AP and BC intersect at point S. Find the ratio BS/SD.
 5. Let a_1, b_1, c_1 be natural numbers. We define $a_2 = \gcd(b_1, c_1)$, $b_2 = \gcd(c_1, a_1)$, $c_2 = \gcd(a_1, b_1)$, and $a_3 = \text{lcm}(b_2, c_2)$, $b_3 = \text{lcm}(c_2, a_2)$, $c_3 = \text{lcm}(a_2, b_2)$. Show that $\gcd(b_3, c_3) = a_2$.
 6. Let $P(x) = x^3 + ax^2 + b$ and $Q(x) = x^3 + bx + a$, where a, b are non zero real numbers. Suppose that the roots of the equation $P(x) = 0$ are the reciprocals of the roots of the equation $Q(x) = 0$. Prove that a and b are integers. Find the greatest common divisor of $P(2013!+1)$ and $Q(2013!+1)$.

**